

Flexible Work Arrangements and Firm Outcomes: Experimental Evidence from India

Shreya Chandra*

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Abstract

This paper studies the impact of flexible work arrangements on worker and firm output in the context of a large apparel manufacturing cluster in India. Production is organized as batch processing lines, where tasks are performed sequentially to produce the final product. In the status quo, production line workers (tailors) are hired on piece rate contracts, implying that (1) workers are free to choose their work schedules, resulting in significant dispersion in worker arrival times at the firm, and (2) workers are only incentivized on individual output, as opposed to firm output. I implement an experiment with 60 small factories and 370 workers, where I incentivize all workers in treatment firms to arrive at work by the owners' preferred start time for a period of 4-weeks. Workers are paid a daily bonus equivalent to 15% of average earnings if they arrive on or before the start time. I find that incentives increase the likelihood that workers arrive on or before the preferred start time in treatment firms by 28% relative to control firms. These changes are driven by workers who were consistently late to arrive at the firm during the pre-incentive weeks, and female workers (who were more likely to be late at baseline). There is no effect on worker attendance, daily work hours and the total duration of breaks during the day. These changes in workers' labor supply add up to a significant improvement in worker coordination at the firm level: dispersion in worker arrival times decreases by 16-30% based on several measures of dispersion. I find a statistically significant increase in weekly firm (worker) output by 23% (31%), resulting in an increase in firm revenues (worker earnings) by 31% (23%). Unlike the effects on timely arrival, treatment effects on worker output and earnings are experienced by all workers in the firm, not just by late or female workers. I explore two possible mechanisms behind these effects: (1) treatment changed the task allocation among workers, (2) treatment created positive spillovers to downstream workers in the production line, or both.

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1 Introduction

As countries develop, they undergo structural changes in the organization of labor – in particular, a shift from self-employment to employment in firms (Bandiera et al., 2022). However, the take-up of wage jobs in many low-income contexts still remains low. Moreover, among those who do take-up such jobs, rates of separation and absenteeism remain high, suggesting that the type of wage-jobs offered in such contexts may not match workers’ preferences for either workplace arrangements or the nature of work (Breza and Kaur, 2025). Understanding frictions to wage-employment is thus a crucial topic of research in developing countries. Recent work attributes these low rates of wage-employment to workers’ preferences for flexibility (Cefala et al., 2024; Oh et al., 2024).¹ While flexibility is an important job attribute for workers – providing flexibility may be costly for firms, especially in settings where the production process requires a high degree of coordination among workers (assembly line manufacturing). This creates a trade-off between worker and firm preferences, and may have broader implications for firm growth and economic development if coordination costs increase in scale (Lucas, 1978).

In this paper, I investigate this trade-off between firm and worker preferences for flexible work arrangements, focusing on flexibility in arrival times at work as the main dimension of flexibility. I study the context of a large apparel manufacturing cluster in Tirupur, India which features many small-medium firms employing 5-15 workers (tailors). Firms are organized as batch processing lines, where each worker (tailor) performs a distinct operation on the garment in batches of 80-100 pieces. The status quo work arrangement in this setting reflects the trade-off between firm and worker preferences. Tailors are hired on piece rate contracts and organize as independent contractors in firms. This enables them to choose their work timings, instead of being required to follow uniform work shifts like their counterparts working in large/export firms.² Effectively, this means that workers do not coordinate their work schedules – in data from 60 firms and 370 workers, the difference in arrival times from company’s stated start time ranges from 60 minutes early to 75 minutes late (90-10 gap, see Figure 1). Moreover, because workers are only incentivized on individual output as opposed to team output (e.g. only 100 sleeves + 100 collars versus 100 pieces of the completed garment), workers’ individual choice of work schedules may create a coordination externality on both other workers and the firm owner.

On the other hand, the production technology implies that worker coordination may improve

¹For example, Cefala et al. (2024) document flexible work schedules as the second most important reason (after earnings per day) for choosing a casual job over a regular job.

²In a separate survey of tailors and firm owners in Tirupur, I document large and systematic differences in work arrangements by firm size – see Table A5. Interviews with both small and large firm owners suggests that differences in the provision of flexibility arise from a combination of institutional and technological constraints: (a) labor laws and social compliance from international buyers require large/export firms to systematically record work shifts in order to provide workers government benefits and overtime pay, and (b) large/export firms are more likely to organize production in assembly lines, while small firms follow batch production processes. Although both forms of production involve a high degree of task specialization among workers (i.e. workers perform only 1 distinct task per product/day), assembly lines have a stronger requirement for worker coordination than batch production processes.

production – because operations need to be performed sequentially to produce the final output, the lack of coordination in worker’s arrival times can either (a) limit worker’s ability to specialize on 1 task because they need to fill in for a late worker who usually performs the preceding task, or (b) increase idle time if they need to wait for the preceding task to be completed. Thus, such flexible work arrangements may slow the overall speed of production, and lower productivity by limiting specialization. This is consistent with owner’s beliefs about how flexibility affects production, and why imposing fixed work schedules may increase production: at baseline, owners were asked what type of measures related to their workers would help increase production (un-prompted). 90% of owners reported that they would like their workers to start work at the same time. This far outweighs other measures – 38% of owners would like workers to take fewer/shorter breaks, 25% would like workers to take fewer un-announced leaves and 17% to not bring their children to work (Figure 2a). When asked why coordinated timings would help increase production, 77% of owners chose idle time, and 67% chose task specialization as possible mechanisms (Figure 2b).

I conduct an experiment with 60 firms and 370 workers to test whether flexibility in work schedules affects firm productivity. I incentivize workers in treatment firms to follow fixed work start times for a period of 4-weeks. Treatment incentives are designed as a daily bonus of Rs. 100 (15% of average daily earnings), paid conditional on arriving at the firm by the owners’ pre-specified work start time (hereafter, preferred start time), elicited at the time of baseline. Thus, the incentive encourages both attendance and before-time arrival. Workers in control firms receive unconditional incentives during the treatment incentive period, matched to a randomly chosen treatment firm each week to address concerns about income effects on worker output.

I collect daily data on worker attendance, arrival times, breaks taken, and production starting 2-weeks before the incentives, and throughout the 4-week incentive period. I define a worker as compliant with the treatment if they are present *and* arrived on or before the preferred start time. First, I estimate the effect of the incentives on worker attendance and compliance. I interpret the treatment effect on worker compliance as an upper-bound estimate of workers’ willingness to pay for workplace flexibility. Second, I study the effect of improved compliance on firm and worker output and (other economic measure). Finally, I empirically test two hypothesized channels of increased firm output: (1) an improvement in the allocation of tasks to workers – either through a decrease in multitasking, or better alignment of tasks to workers’ expertise, and (2) spillovers along the production line to downstream workers.

I find that treatment significantly increases the probability that workers arrive at work at or before owners’ preferred start times as elicited at baseline. This effect is large – the fraction of worker days that workers are present and arrive before the preferred time increases by 12 percentage points, a 28% increase relative to a control mean of 45% before the incentives were active. I find no effect on worker attendance, suggesting that the effect on worker compliance is only driven by changes in workers’ arrival times. Heterogeneity analysis suggests that the effect on compliance is almost entirely driven by workers who consistently arrived after the preferred start time at baseline

(hereafter, “late” workers), and female workers, who are more likely than male workers to be late at baseline. Workers in treatment firms do not reduce total daily work hours or increase the number and total duration of breaks to compensate for arriving early. Overall, these changes in worker compliance reduce dispersion in worker arrival times at the firm level - total dispersion in worker arrival times, measured as the difference in arrival times between the earliest and the last worker, decreases by 20 minutes, or 16% relative to the control mean. The deviation from preferred start time for the 90th percentile worker - which indicates the 2nd to last worker to arrive given firms have an average of 6 tailors - decreases by 30% relative to the control mean. Thus, incentives are successful in improving coordination in worker arrival times in treatment firms.

Next, I estimate the intent-to-treat effects of the incentive on worker output and earnings. I find that weekly worker output increases by 0.29 standard deviations (31% relative to the control mean, equivalent to 600-1,100 more pieces per week) and is statistically significant at the 5% level. Weekly worker earnings (*excluding* the incentive amount) increase by 0.21 standard deviations (23% relative to the control mean) and is weakly statistically significant at the 10% level. The magnitude of the earnings increase is equivalent to INR 700 – larger than the weekly incentive amount offered to workers in treatment firms.

Surprisingly, unlike the results on labor supply, increases in weekly worker output and earnings are realized by all workers, and not driven by female or late workers. I explore two channels which can explain these results: first, by changing worker arrival times (and consequently, both the order in which workers arrive *and* better coordination in worker arrival times), treatment may have resulted in a change in task allocation between workers in the firm. This may result in an increase in the extent of task specialization, or a better assignment of tasks to workers’ expertise. Second, the fact that workers for whom incentives were not binding also benefit from incentives suggests the presence of coordination externalities along the production line. For example, increased compliance among late and/or female workers increases their own output each day, but may have spillover benefits to downstream workers (workers who perform the task next in the order). The firm may realize increases in production through the production line even without changes in the task allocation, or changes in the degree of task specialization.³

Finally, I estimate the effect of incentives on overall firm output, measured as the total number of finished pieces produced by all workers in the firm (worker output is the sum of total pieces produced by a worker, across all tasks/products, regardless of whether the garment has been fully stitched). Consistent with the increase in weekly worker output, I find a statistically significant, 0.19 standard deviation increase in weekly firm output – 23% increase relative to the control mean of 2,700 and control standard deviation of 3,300, equivalent to an increase in total weekly output by 600 pieces. Weekly firm revenues increase by 0.18 standard deviations (control mean of INR 47,000, control SD of INR 79,000), and the treatment effect is statistically significant at the 10% level. This translates to a 31% increase in firm revenues in treatment firms during the incentive period.

³Results exploring these hypotheses are work in progress and will be updated in future drafts of the paper.

Finally, weekly labor costs increase by 0.17 standard deviations in treatment firms, translating to a 25% increase relative to the control mean of INR 16,000, control SD of INR 24,000. Assuming that other non-labor costs did not change differentially between control and treatment firms, these results provide suggestive evidence that improving worker coordination increases firm output and profits.

This paper makes several contributions to the literature. The findings of this paper highlight an important, yet under-explored relationship between the provision of flexible work arrangements, and labor supply preferences (Mas and Pallais, 2020). I experimentally estimate workers’ demand for flexible work arrangements and highlight the trade-off between worker preferences for flexibility and firm output. Workers’ demand for flexibility has been documented across several different contexts (Mas and Pallais, 2017; Ho et al., 2024; Baysan et al., 2024; Wiswall and Zafar, 2018). This paper’s main contribution is to link worker’s WTP for flexibility with firm outcomes. I show that while workers individually prefer flexible work arrangements (as demonstrated by (a) the status quo dispersion in arrival times and (b) heterogeneity in response to the incentives), on average, all workers in the firm experience an increase in output and earnings when they arrive by the preferred start time. These findings are consistent with the idea that despite fixed-schedules being a dis-amenity for workers individually, there are potential gains to themselves *and* the firm.

This paper also speaks to a large literature on barriers to firm growth in developing countries. I build on previous work documenting search and hiring frictions in low-income countries (De Mel et al., 2019; Hardy and McCasland, 2023; Bassi and Nansamba, 2022; Caria and Falco, 2024; Carranza et al., 2020) by documenting that even conditional on forming matches, firms may face barriers to coordinating workers. This could have wider economic consequences on firms’ ability to scale their workforce as coordination becomes costlier with scale (Lucas, 1978). My results suggest that improving coordination in worker arrival times at the firm do increase firm output and revenues, suggesting that frictions to coordinating workers can pose a significant barrier to firm growth in developing countries.

I contribute to the growing literature on the internal organization of small, informal firms in developing countries, building on recent work such as Bassi et al. (2022, 2023). I document worker preferences as the source of coordination costs for SMEs in developing countries, and how they affect firms’ ability to specialize.

The remainder of the paper is organized as follows: [section 2](#) describes the study context and sample firms, [section 3](#) describes the experimental design, implementation and data collection, and [section 4](#) lays out the empirical strategy. [section 5](#) presents the main results.

2 Study Context and Sample

The context of my study is Tiruppur district in the state of Tamil Nadu, India. This region is part of a major garment manufacturing hub in India, catering to both export and domestic markets since the 1970s. The production process in Tiruppur is highly decentralized and carried out by sub-contracted units, called job-work units – these units will form my study sample. Job work units are *small* factories, employing about 10-30 workers, at least 50% of whom are women.⁴

Firms are organized as batch processing lines, where each worker performs a distinct operation on the garment in batches of 80-100 pieces. Operations need to be performed sequentially in order to produce a finish piece of garment. Firms typically specialize in broad product categories - such as t-shirts, bottomwear, innerwear, or kids wear. The average weekly order size in these firms is about 4,000 pieces, varying with product type.

The study takes place in 4 neighborhoods located in Tirupur North sub-district, Tirupur, Tamil Nadu. These neighborhoods were identified as high density clusters of small-medium job-work units during prior scoping work and area mapping exercises. I used the following criteria to screen firms to be part of the sample:

1. Has a stitching section
2. Employs at least 2 tailors who are not related to the firm owner or who work for pay (piece rate contract)
3. Must deliver finished products back to the buyer/retailer
4. Must receive cut-pieces of cloth to be stitched (ie does not only produce own designs)

All firms that satisfy these criteria were eligible for baseline. In case eligible firms were located adjacent to each other, I approached alternate firms in order to minimize spillovers between treatment and control firms located next to each other.

3 Experiment Design

Firms are randomly assigned to one of two treatment groups - (T) Treatment ($n = 30$), and (C) Control ($n = 30$). The treatment group was further randomized into (T1) high and (T2) low incentive sub-treatments, with daily bonus amounts set at Rs. 100 (high) and Rs. 50 (low) respectively. These amounts are approximately 15% and 7.5% of worker's average daily earnings and were chosen based on owners' estimates of the minimum amount needed to encourage workers to come to work early. Workers' average daily earnings are Rs. 600. Treatment incentives are

⁴Note that these firms are of course larger than the typical small developing country firm; however they employ a substantial fraction of the workforce within the textile industry in Tiruppur district - as of 2013, the share of total employment in textiles in 10-30 worker firms ranged between 20-30% in rural and urban areas in Tiruppur district (source: Economic Census, 2013-14). Such organization structures are common in many industrial clusters around the world – see Nadvi and Schmitz (1994); Banerjee and Munshi (2004); Chaudhry and Woodruff (2013); Atkin et al. (2017).

conditional on the worker arriving at the firm and starting work by the company’s preferred start time. Thus, the incentive encourages both on-time arrival and attendance. The total incentive amount was paid to workers as part of their weekly earnings.⁵ The unconditional incentive amount for control group firms was matched to the total realized payment made to a randomly selected treatment firm, capped at Rs. 300 per week (the maximum incentive amount in the low incentive treatment group T2). These treatment-control firm (randomized) pairings were created separately each week. [Figure 4](#) shows the experiment design.

3.1 Data and Summary Statistics

Baseline: I conduct a baseline survey with owners to capture owner and firm characteristics, and a complete worker roster, which is followed by a baseline survey of all workers at the firm. Randomization and treatment assignment was done after owner and worker baselines.

Baseline firm and worker characteristics, and balance by treatment arms are reported in [Table 1](#) and [Table 2](#) respectively. On average, firms hire 6 tailors (excluding owners/co-owners), and have been operating for 5-7 years. Firms produce 4,120 pieces of output each week, and earned a profit of INR 43,000 in the last 30 days (USD 490). There are some notable differences between control and treatment firms at baseline – for instance, treatment firms less likely to produce their own order, and export orders, and earn INR 2,300 less in revenues per worker (INR 1300 less in profits per worker). However, all differences, except the probability of producing own designs are statistically insignificant.

Owners directly supervise workers (that is, without a manager/contrator) in almost all firms – owners also perform stitching work in 90% of firms, and work as regular tailors in about half of firms. About 76% owners are male, 52% have completed secondary school and have 19 years of experience in the garment industry.

Workers are on average 37 years old (similar to owners), 44% are male and most workers are married (93%) and have a child younger than 14 years of age (86%). 84% workers live within a 15 minute commute to the firm, have worked at the current firm for 3 years and have 15 years of experience in the garment industry. Worker characteristics are balanced across treatment and control firms ([Table 2](#)).

Order, Task and Piece Rates Data: At baseline, I also record a roster of all styles that the firm is currently producing. For each style, I record the piece rate that the owner receives for a finished stitched piece and its task structure – a list of all tasks which they need to be performed to produce the finished product, the order in which they are performed, and their piece rates. This roster is updated throughout the 6-week study period to incorporate new styles that the firm is producing.

⁵All workers who were working full-time at the firm at any point during the study were eligible to receive incentives. Workers’ full-time status was determined by the owner and verified by the research team using daily attendance data.

Daily Production and Attendance Data: Starting 1-2 weeks after baseline, I collect data *daily* for a period of 6-weeks, out of which workers are paid incentives for 4-weeks.⁶ Each day, I record worker’s attendance, arrival time, break times during the day, and daily output (by product-task). Arrival times and attendance are recorded by surveyor observation and are not prone to mis-reporting in order to game the incentive; remaining data is self-reported by workers. Daily output is recorded for all production workers, which includes the owner and/or co-owner in some firms. I use daily output data from all production workers to construct an independent measure of total firm production. In addition to tailoring/production output, I also record whether the worker did any other work at the firm, such as helper work, receiving or delivering raw material/finished pieces, machine repair, purchasing inputs, etc.

Table A1 summarizes key labor supply and production outcomes during the pre-incentive weeks, and differences across two worker types – gender, and whether the worker consistently arrived after the preferred start time during the pre-incentives weeks (hereafter, referred to as “late” versus “early” workers). Column (1) reports summary statistics for the overall worker sample; columns (2)-(3) report summary statistics for female and male workers, and columns (4)-(6) report summary statistics for early, “late”, and “very late” workers. “Late” and “very late” are indicators for workers whose average pre-incentive arrival times were between 0 and 30 minutes, and above 30 minutes of the preferred start time, respectively.

Female and male workers have similar demographics – the only exceptions being that women are more likely to be married, live closer to the firm, and have 5 years less experience in the industry than men. On the other hand, late and very late workers are more likely to be female, married, and have more children below 14 years of age. There are no differences in commute times between early, late and very late workers, suggesting that arrival times observed in status quo are not driven by commuting differences, but are likely driven by time commitments in the household before the start of the work day.

Pre-incentive attendance rates are similar across men and women, however, there are several notable gender differences in terms of arrival times and compliance: women arrived 21 minutes after the preferred start time on average, while men arrive 11 minutes before the start time – consequently, women were more likely than men to be “late” and “very late”, and 23 percentage points less likely to comply with the preferred start-time - one-third of men’s average compliance rate. On the other hand, attendance rates are the highest among late workers at 83%, 10 percentage points higher than attendance rates among early and very late workers.⁷ More surprisingly, only

⁶In Phase 1, I collected data 1 week pre-incentives and 1 week post-incentives. In Phase 2, I collected 2 weeks of pre-incentives data only, dropping the post-incentive data collection.

⁷This implies an overall average absenteeism rate of 24%, and 17% among late workers. This range is similar to absenteeism rates documented in other studies. Notably, absenteeism rates among late workers are similar to those reported in large, export factories – 11% in Tanzania (Ho and Huang, 2024), and 15% in India (Adhvaryu et al., 2022). The average absenteeism is similar to those observed in casual labor markets, which are closer to the labor market in this study (25% in rural casual labor markets in Burundi (Cefala et al., 2024), and 20-30% in urban casual labor markets in India (Cefala et al., 2024).

65% of early workers, 45% of late workers, and 10% of very late workers complied with owners' preferred start times before the incentives – suggesting that (a) early workers are not the most reliable workers, and (b) late workers do comply almost half of the time. Given these patterns, I expect the incentive to encourage higher attendance rates among early workers, and to encourage both attendance and on-time arrival among late and very late workers. These patterns also suggest a possible trade-off between regular attendance and on-time arrival. Finally, I observe that women seem to compensate for their late arrival by taking fewer number of breaks during the day, so that average daily hours are only 40 minutes (0.6 hours) lower than men. Late workers work 40 minutes more than both early and very late workers, with no striking differences in the number or duration of breaks taken.

In terms of production outcomes, workers perform 1-2 tasks per day on average, and work on non-tailoring tasks 12% of the time. The average piece rate of tasks performed in a day is 80 paise, with some notable differences by gender – women work on tasks with 30 paise lower average piece rates than men; and baseline lateness – average piece rates decline monotonically with baseline lateness. All workers produce 100 pieces per hours, resulting in small daily earnings differences – women earn Rs. 150 less per day than men, and very late workers earn Rs. 200 less per day than early workers. These differences in piece rates and earnings partly stem from the type of tasks performed – men, and early workers are most likely to perform Complex Singer tasks before the incentives. These tasks are slower and have higher piece rates than the average task.⁸

Deliveries Data: At the end of each week, I record all orders received and delivered by the firm in the past week. I use information on total quantity delivered in a week as a separate measure of firm output and sales in that week.⁹ In addition to information on deliveries, owners are also asked to match each delivery to the date that material was received – this allows me to identify distinct “orders” received by the firm. Additionally, for all new materials received, I elicit owners' estimated planned delivery date, which I can compare with the actual delivery dates reported by owners to assess how treatment affects owners' planning of production.

Endline: Finally, endline surveys conducted during the last week of the study capture firm owner's and worker's perceptions of the treatment (in treatment firms), preferences for flexible work arrangements, changes in firm operations during the study and time-use during and outside of work (relative to the pre-study period). These questions allow me to examine (1) the nature of workers' commitments/restraints that drive their preference for flexibility, (2) how compliance with the incentive impacted these commitments/restraints, and (3) whether compliance shifted workers' preferences for flexible (versus strict) work arrangements.

⁸About 50% of all singer tasks are complex tasks, and output per hour on Complex Singer tasks is half of that on other tasks.

⁹As discussed later in the results section, I use firm output constructed from production data as my preferred measure of firm output, since the quantity of deliveries may be affected by stock of finished pieces at the firm which I do not observe at the start of 6-weeks of daily data collection.

3.2 Implementation

The experiment was implemented in two phases - Phase 1 was implemented with 17 firms (9 treatment, 8 control) from November 2024 to January 2025. Phase 2 was implemented with 43 firms (22 treatment, 21 control) from March 2025 to May 2025. [Figure 5](#) shows the study timeline. Phase 1 only included high treatment firms (T1), and the half of the control firms did not receive any incentive (pure-control). In Phase 2, all control firms receive unconditional incentives matched to a randomly chosen treatment firm, up to the maximum allowable amount in low-treatment firms (Rs. 300). These changes were made to simplify field logistics by minimizing both distance between sample firms as well as firm attrition.

Randomization for Phase 1 was conducted by generating a random order of treatment and control assignments, which firms were matched to in the order in which their baseline survey was conducted. Randomization for Phase 2 was stratified on the degree of dispersion in worker arrival times around the owner’s preferred work start time, using worker’s usual arrival times as reported by the owner at baseline. All specifications reported in the paper control for phase by strata fixed effects to account for the method of randomization in each phase ([Bruhn and McKenzie, 2009](#)).

4 Empirical Strategy

I estimate the effect of incentives on worker level outcomes using the following specification:

$$Y_{i,f,t} = \alpha + \beta T_{i,f} \times \mathbb{I}\{\text{During}\}_t + \lambda_{i,f} + \mu_w + \mu_{dow} + \varepsilon_{i,f,t} \quad (1)$$

where $Y_{i,f,t}$ is the outcome of interest for worker i in firm f at time t , and time is either day or week, depending on the outcome variable. $T_{i,f}$ is an indicator that equals 1 if a firm f is assigned to treatment (fixed over time), $\mathbb{I}\{\text{During}\}_t$ is an indicator variable that takes the value 1 if the study week is an incentive week, and $\lambda_{i,f}$ denotes worker fixed effects. μ_w are study-week fixed effects,¹⁰ and μ_{dow} are day-of-week fixed effects (included only in day-level specifications). β is the coefficient of interest, and standard errors $\varepsilon_{i,f,t}$ are clustered at the firm level. In the current analysis, I pool both high and low treatment arms.

I use a similar specification for firm-level outcomes:

$$Y_{f,t} = \alpha + \beta T_f \times \mathbb{I}\{\text{During}\}_t + \lambda_f + \mu_w + \mu_{dow} + \varepsilon_{f,t} \quad (2)$$

¹⁰Study weeks take the value 1 to 4 for the 4 incentive weeks, 0 for the week prior to incentives. Weeks 0-4 are common across Phase 1 and 2. The first pre-incentive week in Phase 2 is coded week -1, and the final, post-incentive week in Phase 1 is coded week 5. In the current analysis, I drop the post-incentives week (week 5).

where indices and variables are defined as above, and standard errors $\varepsilon_{f,t}$ are clustered at the firm level. Some firm outcomes are only measured at endline, in which case I will run [Equation 2](#) without the t subscript and without firm and study-week fixed effects. Day-of-week fixed effects are only included in firm-day level regressions.

5 Results

5.1 Labor Supply

First, I test the effect of the incentives on worker attendance and compliance with work start times. Columns (1) and (2) in [Table 3](#) report results for worker attendance and compliance with the treatment incentives. Compliance is defined as an indicator for being present at the firm, *and* arriving by the owner’s preferred start time. I find no effect on attendance (column (1)), and an 12 percentage point increase in compliance (column (2)) in treatment firms during the incentives weeks. This represents a 28% increase in compliance relative to the pre-incentive control mean (45%), and is driven by workers arriving on time, rather than an a change in worker attendance. The effect on compliance is larger, and more meaningful when conditioned on workers being present at the firm – on average, 59% workers - or 3-4 out of 6 workers - arrived on time conditional on being present at the firm. Treatment incentives increase worker compliance to 76% – this amounts to 4-5 out of 6 workers who arrive before the preferred time (conditional on being present). Given the small size of the workforce in these firms, the treatment encourages almost all workers to arrive before the preferred start time.

Next, I examine whether these effects vary by gender, and baseline lateness in [Table 4](#). Treatment effects on attendance for women and “late” workers are no different than their respective base categories, consistent with the result above that incentives did not change worker attendance in treatment firms. Surprisingly, attendance among early workers does not improve, given that these workers were in effect only incentivized on attendance, and had the lowest attendance rates in the pre-incentives weeks ([Table A1](#)).¹¹ On the other hand, compliance is driven entirely by female workers, and workers who were consistently late at baseline – female workers are 13 percentage points more likely than male workers to comply (column 3); “late” and “very late” workers are 20 and 35 percentage points respectively more likely than “early” workers to comply (column 4). This is expected, since the incentives were more binding on workers who consistently arrived after the preferred start-time, which was more common among women. In fact, this suggests that incentives may reduce the gap in work hours, and potentially earnings between men and women reported

¹¹37% of worker absences are because of social functions, followed by 19% because there was no work at the firm, 18% because either the worker or a family member was sick, and 10% because of local temple/religious festivals. A majority of these reasons relate to importance of social/kinship relations, as emphasized in [Breza and Kaur \(2025\)](#), but it is worth noting a large fraction of absenteeism attributable to low labor demand. This matches the rate of all-day firm closures recorded in the data - 10% of firms were closed for the entire day. 40% of closures were because the firm did not have work, followed by 36% because of local temple festivals or the firm owner’s personal commitments, and 10% because of power cuts.

earlier (Table A1).

Do workers compensate for arriving earlier at the firm by working fewer hours, or taking more breaks during the work day? Table 5 reports treatment effects on daily work hours, the number and duration of breaks taken in a day. Daily work hours is defined as the total hours worked in a day, excluding the total time spent on breaks. Breaks include any time taken for lunch, tea, or other reasons for which the worker left the firm.¹² All outcomes are coded 0 on days that a worker was absent. Workers in treatment firms work 0.03 additional hours, or 1.8 additional minutes, relative to workers in control firms (column (1)), who work 6.7 hours per day on average pre-incentives. Similarly, I do not find any statistically or economically significant changes in the number of breaks (column (2)) or total duration of breaks (column (3)).

The treatment effects on work hours and break duration vary slightly across male and female workers, and “early” versus “late” workers; however, most of these differences are small in magnitude and statistically insignificant, with the exception of “very late” workers, who increase the total number of breaks taken in a day by 0.21 breaks (statistically significant at the 10% level). This is not surprising when looking at the treatment effect on deviation from the preferred start time, reported in Table A2. On average, workers in treatment firms arrived 10 minutes earlier relative to control workers, but this effect is driven by female (column (2)), and “late” and “very late” workers (column (3)). In fact, “early” workers arrived 10 minutes *after* the preferred start time, while “late” and “very late” workers arrived 8 and 39 minutes earlier relative to control workers.

5.1.1 Does treatment improve worker coordination in arrival times?

Next, I examine how treatment affects worker coordination at the *firm-day level*, conditional on being present. Table A3 reports treatment effects for three measures of dispersion in worker arrival times at the firm-day level: minutes between the 1st and last worker to arrive at the firm (column (1)), the 90-10 gap in worker arrival times (relative to the preferred start time) (column (2)), and the deviation from preferred start time for the 90th percentile worker (columns (3)). This regression excludes firm-days on which only 1 worker was present at the firm, since the notion of coordination is not meaningful in such cases.

Total dispersion in worker arrival times is 131 minutes on average at baseline – treatment reduces dispersion by 20 minutes, a 16% reduction in dispersion, or improvement in coordination relative to control firms (column (1)). The 90-10 gap is a more conservative measure of dispersion which excludes extreme outliers – workers who arrive extremely early or extremely late. For example, the 90-10 gap is 113 minutes – 20 minutes lower than total dispersion in control firms during the pre-incentives week(s). Treatment further reduces the 90-10 gap by 18 minutes, a 16% decrease relative to the control mean of 113 minutes pre-incentives. However, this effect is not significant (column (2)). Finally, I look at treatment effects on deviation from the owner’s preferred start time for the

¹²Workers rarely perform other work for the firm during breaks (5% of worker-days).

90th percentile worker. This measure allows me to test whether these effects are driven by workers for whom the incentive was binding. Indeed, the deviation of the 90th percentile work reduces by 21 minutes in treatment firms, relative to the control group mean of 71 minutes pre-incentives. These effects amount to a 30% reduction in dispersion, or improvement in coordination as a result of treatment.

5.2 Worker Output and Earnings

Next, I test whether incentives increase worker output and earnings. Worker output is measured as the total quantity of pieces stitched by the worker in a week, across all tasks (and products). Note that this does not necessarily correspond to firm output, since a worker who stitched 100 collars and 100 sleeves would be considered to have worked on 200 pieces on that day. However, this measure aligns best with workers earnings, which are constructed as the product of the task specific piece rate and quantity produced on that task, summed over all tasks and products worked on in a week. Output and earnings are coded as 0 on days that a worker was absent, before aggregating to the weekly level. Since firms produce different types of products, I normalize weekly worker output and earnings to pre-incentive, product level, weekly means and standard deviations.

Worker Output Results for weekly worker output are reported in columns (1)-(3) of [Table 7](#). The treatment effect on weekly worker output is 0.29 standard deviations and statistically significant at the 5% level. This amounts to a 31% increase in weekly output relative to the control group mean (column (1) of [Table 7](#)). The effects on weekly output are not statistically different among females, or “late” and “very late ” workers compared to their male/early counterparts (columns (2)-(3) in [Table 7](#)). These results look similar when I use actual levels of output by product type, including product fixed effects to control for level differences by product type (see [Table A4](#)). In levels, workers in treatment firms produced on average 600-1,100 additional pieces per week relative to the average of 3,300-3,700 in control firms, depending on how I aggregate output over different product types.

Worker Earnings Results for weekly worker earnings are reported in columns (4)-(6) of [Table 7](#). The treatment effect on worker earnings mirrors the results on worker output, albeit are smaller in magnitude. This is consistent with the average piece rate per task of INR 0.87. Weekly earnings increase by 0.21 standard deviations (23% of the control mean), but this effect is weakly significant at the 10% level. Notably, treatment partly closes the weekly earnings gap between male and female and early and very late workers in pre-incentive weeks, as documented in [Table A1](#).

Surprisingly, and unlike the results on worker compliance, worker output and earnings are not only concentrated among either female or late and very late workers. This may be due to several reasons, which will determine further analysis and mechanism tests in the paper. Specifically, I hypothesize two reasons explaining the result on worker earnings: first, by changing worker arrival times (and consequently, both better coordination in arrival times, and a different order in which

workers arrive), treatment may have resulted in a change in task allocation between workers in the firm. It is an open question whether the new task allocation is more efficient for workers and firms. I plan to investigate this further by exploring whether new task allocations are better aligned with worker’s expertise¹³, or increase the degree of task specialization in the firm. Second, the fact that workers for whom incentives were not binding also benefit from incentives suggests the presence of coordination externalities along the production. For example, increased compliance among late and/or female workers increases their own output each day, but may have spillover benefits to downstream workers (workers who perform the task next in the order). The firm may realize increases in production through the production line even without changes in the task allocation, or changes in the degree of task specialization. These results are work-in-progress and will be included in future revisions of the paper.

5.3 Total Firm Output and Revenue

How do the changes in individual worker output add up to total firm output and revenues? I construct total firm output using daily production data collected from workers. Total firm output is constructed by adding up total quantity produced on the last task in the production line, across all workers, for each distinct product. Total output is summed by broad product category over the week, and normalized to pre-incentive, product level, weekly means and standard deviations.

I calculate weekly firm revenue as the product of total firm output and the total piece rate that owners earn, and weekly labor cost as the product of total firm output and the total piece rate that workers earn (across the product’s constituent tasks) in each week. Finally, to absorb week by week variation and improve the precision of my estimates, I average total weekly production, revenues and labor costs across the 2 pre-incentive weeks (before) and 4 during-incentive weeks (during), and estimate a double-difference version of [Equation 2](#).

[Table 8](#) reports results on weekly firm output (column (1)), revenue (column (2)) and total labor cost (column (3)).

I find a 0.19 standard deviation increase in average weekly output in treatment firms during the incentive weeks – relative to the control mean and standard deviation, this is a 23% increase in total weekly firm output (control mean = 2,700). This effect is statistically significant at the 10% level, and is consistent with the magnitude of the effect on individual worker output. Weekly firm revenues increase by 0.18 standard deviations – a 31% increase relative to the control group (control mean = INR 47,000 / USD 540). This effect is significant at the 5% level. Finally, total labor costs to the firm increase by 0.17 standard deviations – 25% relative to the control group

¹³Every week, I record worker productivity at the task they are performing at that time. Productivity is measured as the number of pieces of that task completed within 5-minutes and is recorded by surveyor observation. I also elicited which tasks workers consider themselves to be an expert at in worker baseline surveys. I plan to use these data to assess whether treatment improves task allocation by making workers more likely to work on their expert tasks.

(control mean = INR 16,000 / USD 183), and marginally significant at the 10% level ($p\text{-value} = 0.10$). These results provide suggestive evidence that improving worker coordination increases firm output and profits, assuming that other fixed, non-labor costs did not change differentially between control and treatment firms. Further analysis will use self-reported profits, revenues, and non-labor costs from the owner endline survey to comprehensively assess the profitability of incentives, and compare these estimates to total weekly payouts made to workers to encourage them to arrive on time.

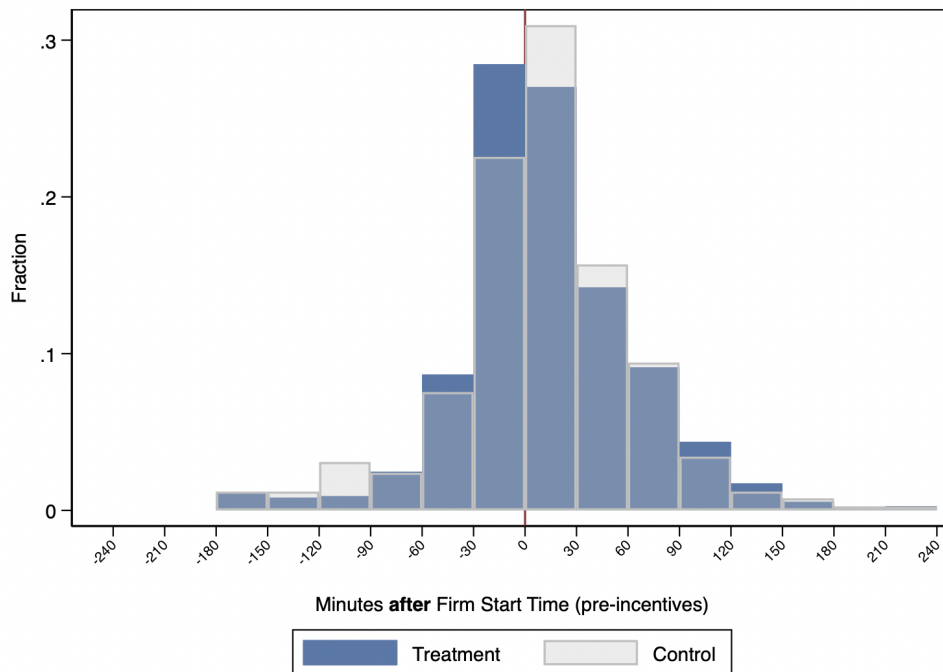
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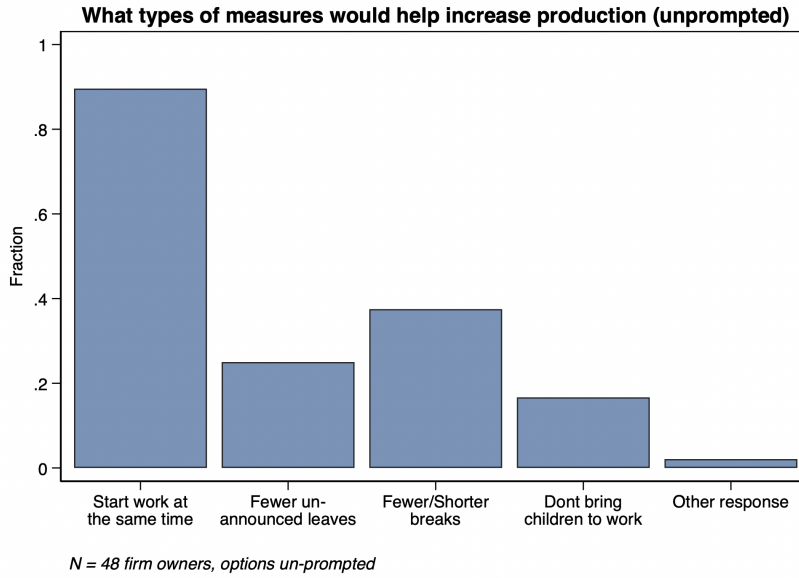
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6 Figures

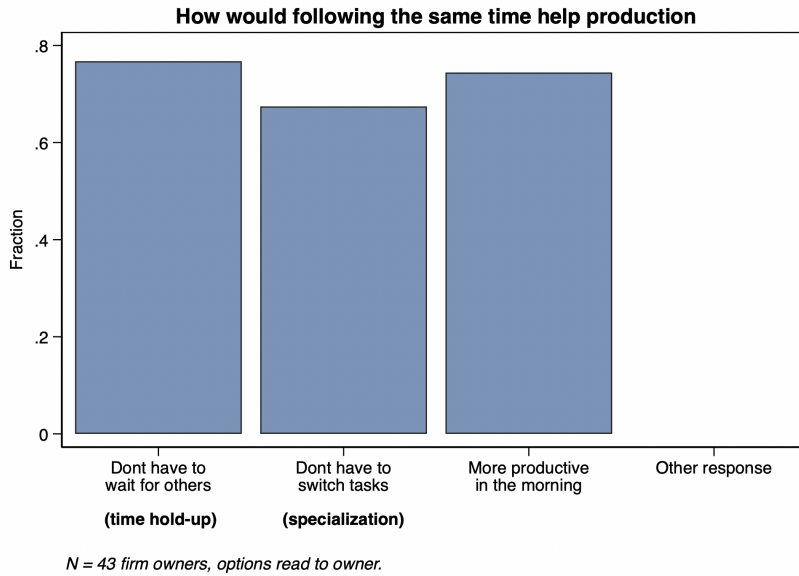
Figure 1: Dispersion in Worker Arrival Times, before Incentives



Notes: This figure plots the distribution of the deviation of worker arrival times from the owner's preferred start time. N = 60 firms, 420 workers.



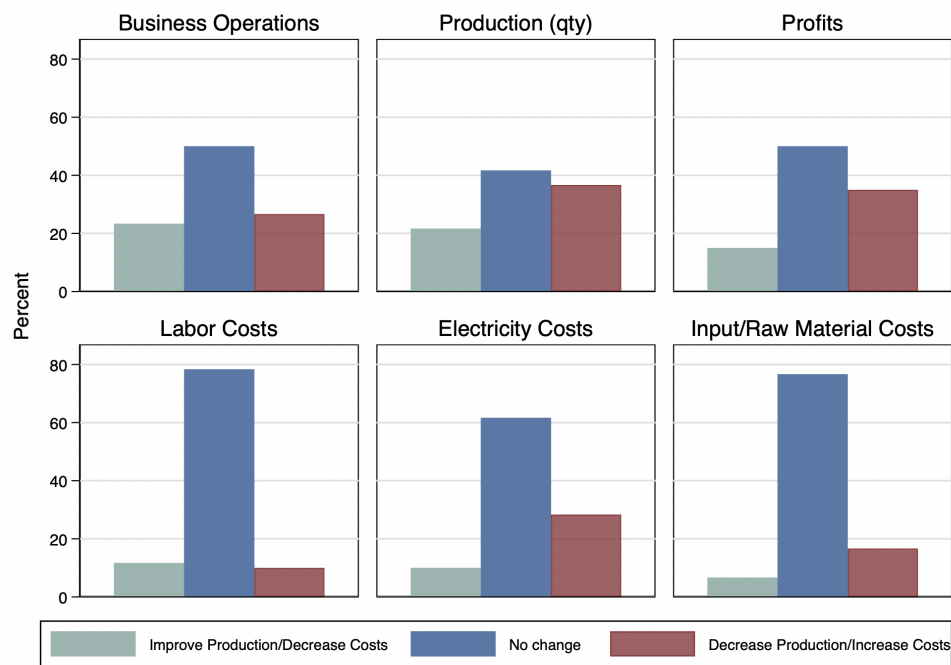
(a) Measures that could help increase production



(b) Why is coordination in worker arrival times important

Notes: Panel A reports owner's responses from the question "What type of measures would help increase production?", with options unprompted. $N = 60$. Panel B reports owner's responses from the question "If everybody starts work at the same time, how will it improve production?", with options 1 = Workers don't have to wait for others, 2 = Workers don't have to switch tasks, 3 = Workers are more productive during morning, and Others. This data is from the full sample of 60 firm owners.

Figure 3: How Providing Flexibility Affects Firm Operations



Notes: This figure reports owner's responses from the question: "Now I will list 4 different business objectives. For each of these, do you think providing workers flexibility affects this objective positively, negatively, or has no effect?" This data is from the full sample of 60 firm owners.

Figure 4: Experiment Design

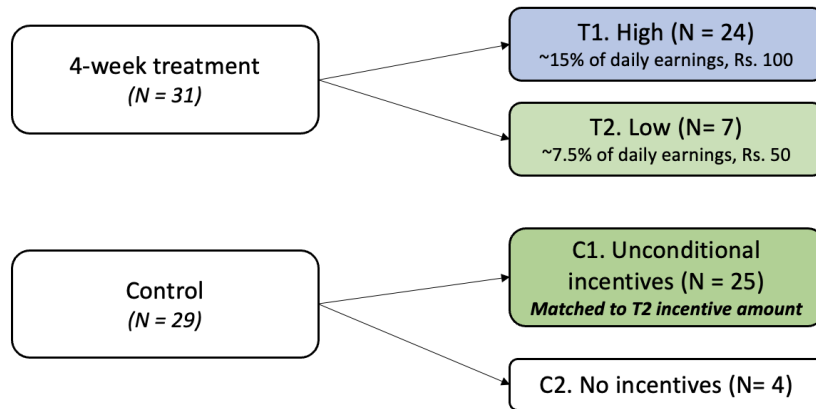


Figure 5: Study Timeline

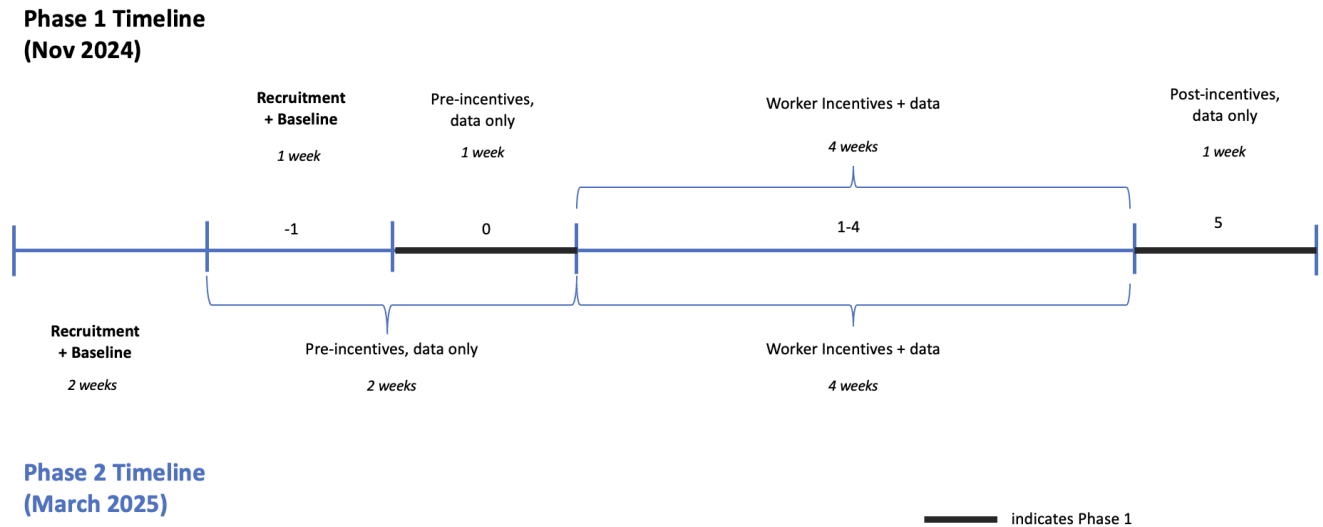
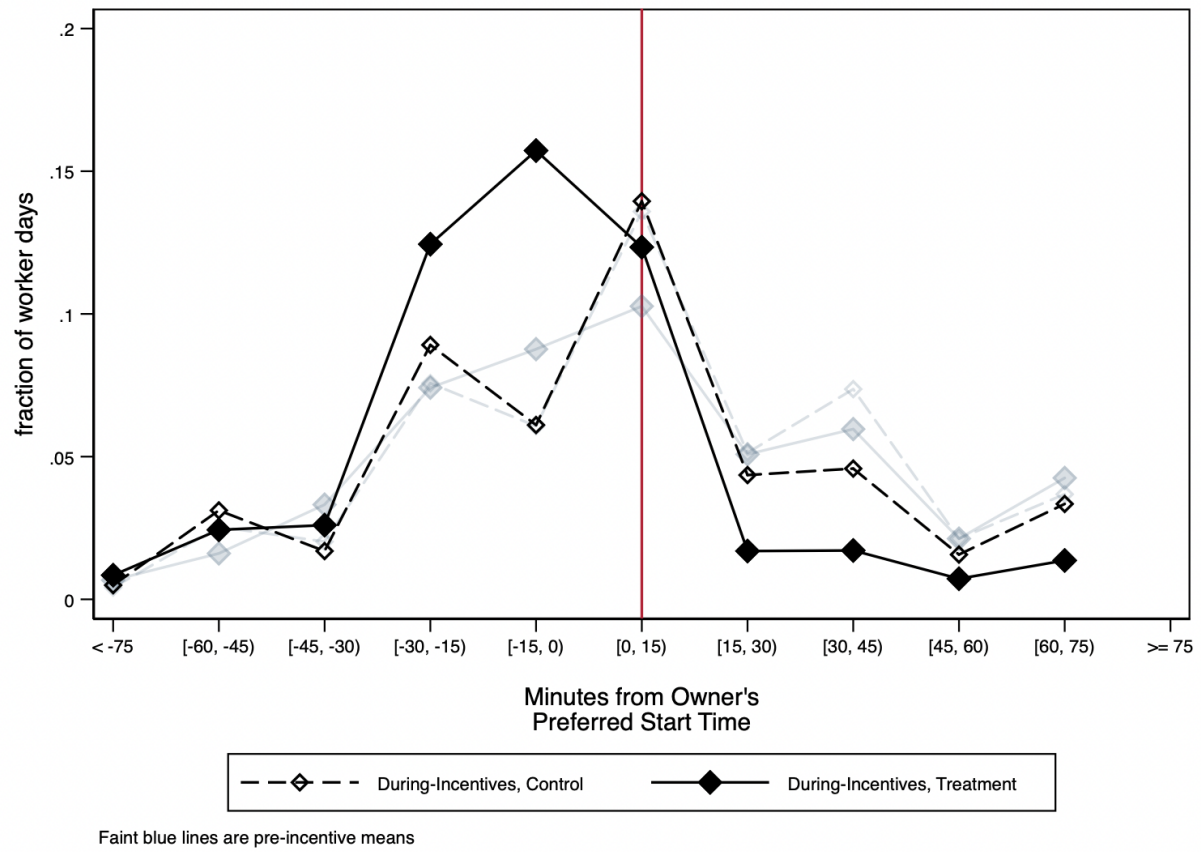


Figure 6: Change in deviation from owner's preferred start time



7 Tables

Table 1: Balance on Baseline Firm Characteristics

	Control	Treatment	T - C	<i>p-value</i>
	Mean/SD	Mean/SD	Difference	
Firm Characteristics				
# Tailors	6.28 (3.22)	5.97 (3.31)	-0.33	0.70
# Helpers	1.79 (1.15)	1.23 (1.06)	-0.58	0.03
Total # Machines	11.38 (4.52)	10.06 (3.03)	-1.35	0.17
Owner directly supervises workers	0.93 (0.26)	1.00 (0.00)	0.07	0.16
Firm Age (yrs)	7.38 (5.85)	5.30 (4.85)	-2.08	0.15
Produce own orders	0.28 (0.45)	0.03 (0.18)	-0.24	0.01
Export orders	0.14 (0.35)	0.06 (0.25)	-0.08	0.35
Weekly Output	4120.00 (2346.45)	4128.57 (3232.34)	96.74	0.89
Revenues (last 30 days)	147855.56 (137491.26)	119066.67 (104577.62)	-29888.47	0.28
Revenue per worker	19495.15 (14922.86)	17236.36 (13134.35)	-2360.59	0.49
Profits (last 30 days)	39459.56 (73012.58)	22633.33 (16643.28)	-17143.33	0.22
Profits per worker	4723.93 (5139.08)	3405.96 (2518.97)	-1342.23	0.20
Total Costs - wage bill (last 30 days)	31025.17 (21543.84)	32591.61 (21720.48)	1496.43	0.79
Owner Demographics				
Age	37.10 (7.87)	35.74 (6.97)	-1.31	0.50
Male	0.76 (0.44)	0.77 (0.43)	0.02	0.89
Completed Secondary School	0.52 (0.51)	0.39 (0.50)	-0.14	0.29
Commute Time (mins)	5.69 (6.33)	8.19 (9.62)	2.46	0.25
Experience in Industry (yrs)	18.83 (10.73)	16.70 (8.30)	-2.08	0.40
Owner does stitching work	0.90 (0.31)	0.81 (0.40)	-0.08	0.29
Owner stitches as a regular tailor	0.45 (0.51)	0.52 (0.51)	0.07	0.51
Observations	29	31		

Notes: This table reports means, mean differences by treatment group and p-values for baseline firm and owner characteristics. Mean differences and p-values are estimated using a regression of each variable on an indicator for treatment with phase x strata fixed effects, with robust standard errors.

Table 2: Balance on Baseline Worker Characteristics

	Control	Treatment	T - C	<i>p-value</i>
	Mean/SD	Mean/SD	Difference	
Age	36.62 (9.15)	36.48 (9.74)	-0.27	0.83
Male	0.44 (0.50)	0.41 (0.49)	-0.03	0.62
Married	0.93 (0.45)	0.91 (0.49)	-0.02	0.69
Has children < 14 yrs	0.86 (1.36)	0.84 (1.30)	-0.04	0.87
Has elderly member in HH	0.37 (0.48)	0.37 (0.49)	0.01	0.87
Related to owner	0.25 (0.43)	0.28 (0.45)	0.03	0.65
Native to TN	0.91 (0.29)	0.96 (0.20)	0.05	0.13
Commute < 15 mins	0.84 (0.37)	0.84 (0.36)	0.01	0.85
Completed Secondary School	0.46 (0.50)	0.40 (0.49)	-0.06	0.27
Experience in Industry (yrs)	14.70 (9.05)	14.34 (9.96)	-0.51	0.67
Tenure at Firm (yrs)	2.95 (4.03)	2.08 (3.07)	-0.88	0.16
Observations	189	187		

Notes: This table reports means, mean differences by treatment group and p-values for baseline firm and owner characteristics. Mean differences and p-values are estimated using a regression of each variable on an indicator for treatment with phase x strata fixed effects, with standard errors clustered at the firm level.

Table 3: Worker Attendance and Compliance

	Attendance	Compliance	Compliance cond. on Attendance
	(1)	(2)	(3)
Treatment $\times$ During Incentives	0.00 [0.04]	0.12*** [0.04]	0.17*** [0.04]
Control Mean (Pre-Incentives)	.75	.45	.59
Control Mean (Incentive Weeks)	.76	.52	.68
Treatment Effect %	0	.28	.29
Observations	10238	10238	7761

Notes: All specifications include worker, day-of-week, week and phase $\times$ strata fixed effects. Standard errors are clustered at the firm level. Attendance is defined as an indicator for whether a worker was present at the firm on that day. Compliance is defined as an indicator for whether a worker was present at the firm, *and* arrived by the owner's preferred start time on a particular day. The sample for these regressions excludes days on which a firm was shut, and no production took place.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 4: Heterogeneity in Treatment Effects on Compliance and Attendance

	Attendance		Compliance	
	(1)	(2)	(3)	(4)
Treatment $\times$ During	0.01 [0.04]	0.00 [0.05]	0.05 [0.05]	-0.04 [0.05]
Treatment $\times$ During $\times$ Female	-0.01 [0.04]		0.13** [0.05]	
Treatment $\times$ During $\times$ Late		-0.00 [0.06]		0.20*** [0.05]
Treatment $\times$ During $\times$ Very Late		0.00 [0.08]		0.35*** [0.08]
Control Mean (Pre-Incentives)	.75	.75	.45	.45
Omitted Group Mean (Pre-Incentives)	.74	.68	.57	.61
Treatment Effect %	.01	0	.11	-.09
<i>p-val</i> T $\times$ During $\times$ Female + T $\times$ During = 0	0.90		0.00	
<i>p-val</i> T $\times$ During $\times$ Late + T $\times$ During = 0		0.96		0.00
<i>p-val</i> T $\times$ During $\times$ Very Late + T $\times$ During = 0		0.98		0.00
Worker FE				
Observations	10238	10238	10238	10238

Notes: All specifications include worker, day-of-week, week, and phase $\times$ strata fixed effects, from the specification in [Equation 1](#). Standard errors are clustered at the firm level. Attendance is defined as an indicator for whether a worker was present at the firm on that day. Compliance is defined as an indicator for whether a worker was present at the firm, *and* arrived by the owner's preferred start time on a particular day. The sample for these regressions excludes days on which a firm was shut, and no production took place.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 5: Treatment Effects on Daily Work Hours

	Work Hours	Total # Breaks	Total Break Duration (mins)
	(1)	(2)	(3)
Treatment $\times$ During Incentives	0.03 [0.39]	0.18 [0.14]	-0.44 [3.45]
Control Mean Pre-Incentives	6.78	2.15	51.34
Control Mean Incentive Weeks	6.96	1.78	54.48
Treatment Effect %	0	.08	-.01
Observations	10036	7818	10238

Notes: All specifications include dow, week and phase $\times$ strata fixed effects. Standard errors are clustered at the firm level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 6: Heterogeneity in Treatment Effects on Daily Work Hours

	Work Hours		Total # Breaks		Total Break Duration (mins)	
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment $\times$ During	-0.13 [0.40]	-0.28 [0.48]	0.13 [0.14]	0.10 [0.13]	0.07 [3.41]	-2.75 [4.16]
Treatment $\times$ During $\times$ Female	0.28 [0.42]		0.09 [0.09]		-0.85 [3.54]	
Treatment $\times$ During $\times$ Late		0.17 [0.57]		0.06 [0.13]		2.81 [4.52]
Treatment $\times$ During $\times$ Very Late		0.85 [0.69]		0.21* [0.12]		4.87 [5.26]
Control Mean (Pre-Incentives)	6.78	6.78	2.15	2.15	51.34	51.34
Omitted Group Mean (Pre-Incentives)	6.92	6.42	2.15	2.33	51.98	47.36
Treatment Effect %	-.02	-.04	.06	.05	0	-.05
<i>p-val</i> T $\times$ During $\times$ Female + T $\times$ During = 0	0.75		0.15		0.85	
<i>p-val</i> T $\times$ During $\times$ Late + T $\times$ During = 0		0.81		0.37		0.99
<i>p-val</i> T $\times$ During $\times$ Very Late + T $\times$ During = 0		0.38		0.05		0.67
Worker FE	X	X	X	X	X	X
Observations	10036	10036	7818	7818	10238	10238

Notes: All specifications include day-of-week, week, phase $\times$ strata and worker fixed effects, from the specification in Equation 1. Standard errors at clustered at the firm level. Work hours are calculated as the difference between worker's exit and entry times, subtracting the total time spent on lunch, tea and other breaks during which the worker was not stitching. Total # of breaks includes the number of lunch, tea and other breaks taken by the worker during the day, and total break duration (in minutes) is the sum of time spent across all work breaks in a day.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 7: Normalized Worker Output and Earnings

	Weekly Output (Qty, normalized)			Weekly Earnings (INR, normalized)		
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment $\times$ During	0.29** [0.13]	0.27** [0.13]	0.12 [0.14]	0.21* [0.11]	0.20 [0.13]	0.17 [0.14]
Treatment $\times$ During $\times$ Female		0.08 [0.10]			0.07 [0.11]	
Treatment $\times$ During $\times$ Late			0.26 [0.16]			-0.00 [0.18]
Treatment $\times$ During $\times$ Very Late			0.37* [0.20]			0.14 [0.23]
Control Mean (Pre-Incentives)	3774.49	3774.49	3774.49	3296.52	3296.52	3296.52
Control SD (Pre-Incentives)	4009.33	4009.33	4009.33	3579.36	3579.36	3579.36
Omitted Group Mean (Pre-Incentives)		4003.25	3613.06		3921.38	3627.02
Treatment Effect %	.31	.28	.13	.23	.22	.19
<i>p-val</i> T $\times$ During $\times$ Female + T $\times$ During = 0		0.01			0.03	
<i>p-val</i> T $\times$ During $\times$ Late + T $\times$ During = 0			0.02			0.29
<i>p-val</i> T $\times$ During $\times$ Very Late + T $\times$ During = 0			0.02			0.13
Worker FE	X	X	X	X	X	X
Observations	1877	1816	1877	1877	1816	1877

Notes: SEs clustered at firm level. All specifications include day-of-week, week and phase $\times$ strata FE. This regression excludes owners/co-owners who also do stitching work at the firm. This regression excludes days of firm closure if no production took place at the firm. Output is normalized by product relative to pre-incentive week means. Workers who are absent on a given day are assumed to have worked on the product that most other workers in the firm worked on. Earnings are coded as 0 for workers who are absent on a given day. Output and earnings are winsorized at the 99th percentile before normalizing.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 8: Treatment Effects on Weekly Firm Output - Pre vs. During Incentives x Treatment

	# Pieces	Revenues (INR)	Total Labor Cost (INR)
	(1)	(2)	(3)
Treatment $\times$ During	0.19* [0.11]	0.18** [0.09]	0.17 [0.10]
Control Mean (Pre-Incentives)	2743.6	46784.16	16406.68
Control SD (Pre-Incentives)	3347.44	79252.26	24136.92
Treatment Effect %	.23	.31	.25
p-value (Treatment $\times$ During)	0.09	0.04	0.10
Firm FE	X	X	X
Observations	139	139	139

Notes: All specifications include firm and phase $\times$ strata fixed effects. Standard errors are clustered at the firm level. Production/Delivery quantity is winsorized at the 99th percentile (by product, phase and incentive week). Revenues and total labor costs are calculated using winsorized quantities. All outcomes are normalized to pre-incentive, product level means to account for level differences in production quantities, and summed to the firm-week level, and then averaged to pre- and during incentive weeks.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

A Appendix Tables

Table A1: Worker Labor Supply and Production Outcomes - Pre Incentives

	All Workers	by Gender		by Avg. Arrival Time before incentives		
	(1)	Female (2)	Male (3)	Early (4)	0-30 mins late (5)	30+ mins late (6)
Demographics						
Female	0.56 (0.50)			0.36 (0.48)	0.64 (0.48)	0.79 (0.40)
Age	37.43 (9.57)	36.38 (7.91)	38.89 (11.31)	37.50 (10.62)	38.01 (8.65)	36.83 (8.72)
Married	0.82 (0.38)	0.90 (0.30)	0.72 (0.45)	0.74 (0.44)	0.86 (0.35)	0.91 (0.29)
# children < 14 yrs	0.89 (1.49)	1.08 (1.78)	0.62 (0.87)	0.64 (0.84)	0.98 (1.75)	1.16 (1.85)
Commute < 15 mins	0.83 (0.37)	0.87 (0.34)	0.79 (0.41)	0.82 (0.39)	0.85 (0.36)	0.85 (0.36)
Experience in Industry (yrs)	15.03 (9.67)	12.60 (8.49)	18.39 (10.19)	15.64 (10.04)	15.50 (9.35)	13.77 (9.30)
Attendance, Arrival Times						
Present	0.76 (0.43)	0.78 (0.41)	0.74 (0.44)	0.71 (0.46)	0.84 (0.36)	0.78 (0.42)
Minutes after owner's preferred start time	13.99 (77.38)	31.27 (79.47)	-9.77 (67.59)	-29.57 (50.04)	14.14 (43.29)	71.43 (92.91)
On-time or Early	0.44 (0.50)	0.29 (0.45)	0.64 (0.48)			
0-30 mins late	0.26 (0.44)	0.29 (0.46)	0.22 (0.41)			
30+ mins late	0.30 (0.46)	0.42 (0.49)	0.14 (0.35)			
Complied with Owner's Preferred Start Time	0.46 (0.50)	0.37 (0.48)	0.57 (0.50)	0.64 (0.48)	0.48 (0.50)	0.17 (0.37)
Total Work Hours	6.94 (4.22)	6.86 (4.01)	7.04 (4.48)	6.81 (4.65)	7.78 (3.68)	6.39 (3.87)
Total # Breaks	2.06 (1.06)	1.99 (1.00)	2.17 (1.12)	2.21 (1.09)	2.08 (1.03)	1.84 (1.00)
Total Break Duration - mins	52.63 (35.61)	52.09 (34.46)	53.33 (37.05)	50.64 (37.34)	58.82 (32.10)	50.19 (35.31)
Production						
# Unique Tasks	1.55 (1.44)	1.63 (1.43)	1.43 (1.45)	1.59 (1.63)	1.66 (1.39)	1.38 (1.13)
Non-tailoring work	0.12 (0.32)	0.11 (0.32)	0.12 (0.33)	0.11 (0.31)	0.16 (0.37)	0.09 (0.29)
Avg Task Piece Rate (INR)	0.87 (1.29)	0.77 (1.03)	1.00 (1.56)	1.00 (1.62)	0.85 (1.14)	0.69 (0.73)
Output per hour	104.35 (95.85)	101.00 (86.55)	108.92 (107.13)	114.13 (118.18)	101.19 (86.58)	94.10 (63.97)
Daily Earnings (INR)	556.15 (672.52)	494.38 (563.89)	635.63 (783.74)	631.75 (818.07)	571.88 (610.81)	430.48 (419.83)
Singer Task	0.17 (0.38)	0.14 (0.34)	0.22 (0.42)	0.20 (0.40)	0.17 (0.38)	0.13 (0.34)
Complex Singer Task	0.09 (0.28)	0.04 (0.20)	0.14 (0.35)	0.14 (0.35)	0.07 (0.25)	0.02 (0.14)
Observations	3030	1705	1325	1339	787	904

Notes: Standard deviations in parentheses. Data is at the worker-day level, including only 1-2 weeks of pre-incentives data from each phase. Work hours exclude the total duration of breaks taken during the day. Output is calculated as the sum of pieces stitched across all tasks and products in a day. Work hours, breaks and all production outcomes are coded as 0 on days that the worker was absent.

Table A2: Treatment Effects on Deviation from Start Time

	Deviation from Start Time (mins)		
	(1)	(2)	(3)
Treatment $\times$ During	-9.72 [5.99]	-3.86 [6.83]	10.23* [5.32]
Treatment $\times$ During $\times$ Female		-9.76* [5.74]	
Treatment $\times$ During $\times$ Late			-17.60*** [4.25]
Treatment $\times$ During $\times$ Very Late			-49.06*** [7.60]
Control Mean (Pre-Incentives)	12.26	12.26	12.26
Omitted Group Mean (Pre-Incentives)		-13.1	-32.03
Treatment Effect %	-.79	-.31	.83
<i>p-val</i> T $\times$ During $\times$ Female + T $\times$ During = 0		0.04	
<i>p-val</i> T $\times$ During $\times$ Late + T $\times$ During = 0			0.21
<i>p-val</i> T $\times$ During $\times$ Very Late + T $\times$ During = 0			0.00
Worker FE			
Observations	7793	7793	7793

Notes: All specifications include day-of-week, week, phase $\times$ strata and worker fixed effects, from the specification in Equation 1. Standard errors at clustered at the firm level. The sample for these regressions excludes days on which a firm was shut, and no production took place.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A3: Treatment Effects on Coordination - Firm Level

	Mins between 1st and last worker (1)	Mins between the 10th and 90th pctl worker (2)	Deviation from start-time: 90th pctl worker (3)
Treatment $\times$ During Incentives	-20.46 [12.96]	-18.46 [11.57]	-21.41* [12.62]
Control Mean Pre-Incentives	131.18	113.24	71.86
Control Mean Incentive Weeks	108.14	93.67	51.06
Treatment Effect %	-.16	-.16	-.3
Observations	1590	1590	1590

Notes: This table reports firm-day level outcomes on coordination in worker arrival times. All specifications include dow, week and phase $\times$ strata fixed effects. Standard errors are clustered at the firm level. All outcome variables are in minutes, and are only computed for firm-days where strictly greater than 1 worker was present.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A4: Worker Output and Earnings

	Weekly Output (Qty)			Weekly Earnings (INR)		
	(1)	(2)	(3)	(4)	(5)	(6)
Treatment $\times$ During	631.03** [246.07]	639.51** [257.69]	398.37 [273.09]	439.74** [190.76]	402.65 [249.48]	491.98** [210.83]
Treatment $\times$ During $\times$ Female		9.95 [192.42]			69.67 [192.66]	
Treatment $\times$ During $\times$ Late			286.22 [322.71]			-277.31 [271.40]
Treatment $\times$ During $\times$ Very Late			587.86 [365.61]			40.03 [269.89]
Control Mean (Pre-Incentives)	3327.67	3327.67	3327.67	2906.28	2906.28	2906.28
Omitted Group Mean (Pre-Incentives)		3644.75	3184.01		3570.21	3196.31
Treatment Effect %	.19	.19	.12	.15	.14	.17
<i>p-val</i> T $\times$ During $\times$ Female + T $\times$ During = 0		0.02			0.02	
<i>p-val</i> T $\times$ During $\times$ Late + T $\times$ During = 0			0.04			0.46
<i>p-val</i> T $\times$ During $\times$ Very Late + T $\times$ During = 0			0.01			0.06
Worker FE	X	X	X	X	X	X
Product FE	X	X	X	X	X	X
Observations	2301	2207	2301	2301	2207	2301

Notes: SEs clustered at firm level. All specifications include worker, product, week, and phase $\times$ strata fixed effects. This regression excludes owners/co-owners who also do stitching work at the firm. This regression excludes days of firm closure if no production took place at the firm. All outcomes are winsorized at the 99th percentile.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A5: Flexibility by Firm Size

	Small Firms		Large Firms	
	Mean	SD	Mean	SD
Timings Same	0.49	(0.50)	0.77	(0.42)
Timings Strict	0.15	(0.36)	0.50	(0.51)
Entry time is recorded	0.03	(0.17)	0.50	(0.51)
Pay cut if late	0.06	(0.23)	0.40	(0.49)
Margin of Lateness				
Acceptable to be 5 minutes late	0.97	(0.17)	0.94	(0.24)
Acceptable to be ≤ 30 minutes late	0.91	(0.29)	0.69	(0.47)
Acceptable to be > 30 minutes late	0.85	(0.36)	0.62	(0.49)
Observations	70		48	

Notes: SD in parenthesis (even columns). Small firms are defined as firms with up to 20 workers, and where the owner is often both a tailor (either full or part-time) and manager of the firm. Large firms include firms with at least 50 workers, where ownership and factory-floor management are separate.